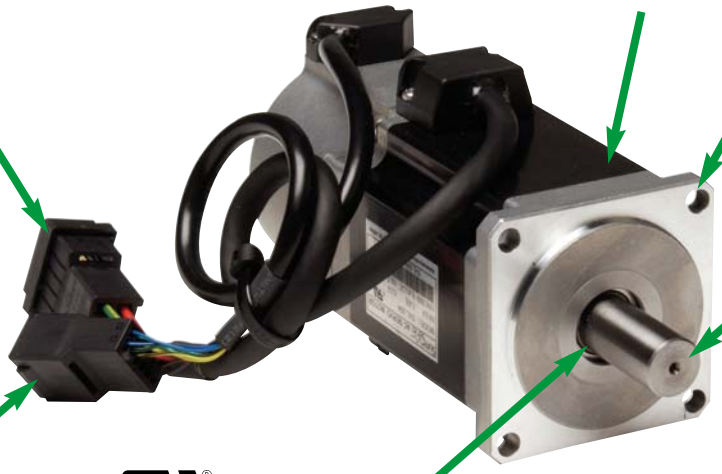
SureServo™ AC Servo Motor Specifications

Servo motor overview

750W and below



Motor Power and Brake Connector
1-meter cable with
6-position connector
(Not liquid tight)

Encoder Connector
1-meter cable with
9-position connector
(Not liquid tight)

IP65 Housing

Low Inertia Motors
- 100W 40 mm flange
- 200W 60 mm flange
- 400W 60 mm flange
- 750W 80 mm flange

Keyless Shafts
- 100W 8 mm diameter
- 200W 14 mm diameter
- 400W 14 mm diameter
- 750W 19 mm diameter

***Without* Shaft Seal**
(not liquid tight)

All SureServo motors have keyless shafts for use with servo-grade clamp or compression couplings.

1 kW and above



Motor Power and Brake Connector
(Liquid tight when using AutomationDirect cables)

Encoder Connector
(Liquid tight when using AutomationDirect cables)

IP65 Housing

Low and Medium Inertia Motors
Low Inertia Model
- 1 kW 100 mm flange

Medium Inertia Models
- 1 kW 130 mm flange
- 2 kW 180 mm flange
- 3 kW 180 mm flange

Keyless Shafts
Low Inertia Model
- 1 kW 22 mm diameter

Medium Inertia Models
- 1 kW 22 mm diameter
- 2 kW 35 mm diameter
- 3 kW 35 mm diameter

***With* Shaft Seal**
(liquid tight)

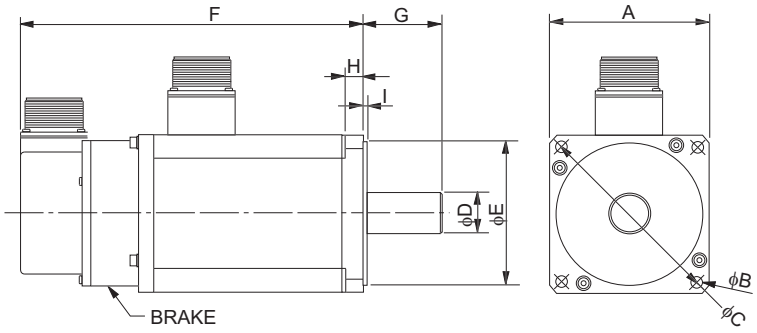
Sure servo™ AC Servo Motor Specifications

Motor Specifications									
Inertia Range		Low					Medium		
Model Name: Sxx-xxx		SVL-201	SVL-202	SVL-204	SVL-207	SVL-210	SVM-210	SVM-220	SVM-230
Price		<--->	<--->	<--->	<--->	<--->	<--->	<--->	<--->
Model with brake: Sxx-xxxB		SVL-201B	SVL-202B	SVL-204B	SVL-207B	SVL-210B	SVM-210B	SVM-220B	SVM-230B
Price		<--->	<--->	<--->	<--->	<--->	<--->	<--->	<--->
Rated output power	**W**	100	200	400	750	1000	1000	2000	3000
Rated torque	**N-m**	0.318	0.64	1.27	2.39	3.3	4.8	9.4	14.3
	lb-in	2.8	5.7	11.2	21.2	29.2	42.5	83.2	125.7
Maximum torque	**N-m**	0.95	1.91	3.82	7.16	9.9	15.7	23.5	35.8
	lb-in	8.4	16.9	33.8	63.4	87.6	138.9	208.0	316.8
Rated speed	**rpm**	3000					2000		
Max. speed	**rpm**	5000			4500		3000		
Rated current	**A**	1.1	1.7	3.3	5.0	6.8	5.6	13.1	17.4
Max. current	**A**	3.0	4.9	9.3	14.1	18.7	17.6	31.4	42.3
Drive input current	**1 phase A**	1.0	1.7	3.4	5.9	8.0	8.0	-	-
	3 phase A	0.8	1.3	2.6	4.7	6.2	6.2	9.1	13.6
Max. radial shaft load	**N**	78.4	196		343	490		784	
	lb	18	44		77	110		176	
Max. thrust shaft load	**N**	39.2	68.6		98			392	
	lb	9	15		22			88	
Brake **Voltage**	**VDC**	24							
Current	**ADC**	0.21	0.38		0.4	0.75	0.83	1.45	1.67
Holding Torque	**N-m**	0.32	1.27		2.55	9.3	7.5	32.0	50.0
	lb-in	2.83	11.24		22.57	82.3	66.38	283.2	442.5
Rotor inertia w/o brake	**kg-m2**	0.03E-4	0.18E-4	0.34E-4	1.08E-4	2.6E-4	5.98E-4	15.8E-4	43.3E-4
	lb-in-s2	0.27E-4	1.59E-4	3.0E-4	9.56E-4	23.0E-4	52.9E-4	139.8E-4	383.2E-4
Rotor inertia with brake	**kg-m2**	0.06E-4	0.28E-4	0.44E-4	1.32E-4	3.1E-4	8.8E-4	27.8E-4	56.3E-4
	lb-in-s2	0.53E-4	2.48E-4	3.9E-4	11.7E-4	27.4E-4	77.9E-4	246.0E-4	498.3E-4
Mechanical time constant	**ms**	0.6	0.9	0.7	0.6	1.7	1.4	1.6	0.9
Static friction torque	**N-m**	0.02	0.04		0.08	0.49	0.29	0.98	
Torque constant-KT	**N-m/A**	0.32	0.39	0.4	0.5	0.56	0.91	0.77	0.86
Voltage constant-KE	**V/rpm**	33.7E-3	41.0E-3	41.6E-3	52.2E-3	58.4E-3	95.71E-3	81.1E-3	90.5E-3
Armature resistance	**Ω**	20.3	7.5	3.1	1.3	2.052	1.98	0.6	0.162
Armature inductance	**mH**	32	24	11	6.3	8.4	13.2	6.1	2.3
Electrical time constant	**ms**	1.6	3.2	3.2	4.8	4.1	6.7	10.1	14.2
Motor Type		Brushless, AC, permanent magnet [Neodymium (Nd), Iron (Fe), Boron (B)]							
Insulation class		Class F							
Insulation resistance		>100 MΩ , 500 VDC							
Insulation strength		1500 VAC, 50 Hz, 60 seconds							
Ambient temperature range		0 to 40°C (32°F to 104°F)							
Operating temperature *(measured case temperature)*		70°C (158°F)							
Maximum operating temperature *(measured case temperature)*		70°C + 40°C = 110°C (230°F)							
Storage temperature		-20 to 65°C (-4 to 149°F)							
Operating humidity		20 to 90% RH (non-condensing)							
Storage humidity		20 to 90% RH (non-condensing)							
Vibration / Shock		2.5G / 5.0G							
Environmental rating		IP65 motor body; IP40 shaft; IP20 connector				IP65 (requires *Sure*Servo cables)			
Weight without brake	**kg**	0.5	0.9	1.3	2.5	4.7	4.8	12.0	17.0
	lb	1.1	1.98	2.87	5.5	10.36	10.58	26.46	37.48
Weight with brake	**kg**	0.7	1.4	1.8	3.4	6.3	7.5	19.0	24.0
	lb	1.54	3.09	3.97	7.5	13.89	16.53	41.89	52.9
Agency Approvals		CE; UL recognized (U.S. and Canada)							

NOTE: U.S. customary units are for reference only.

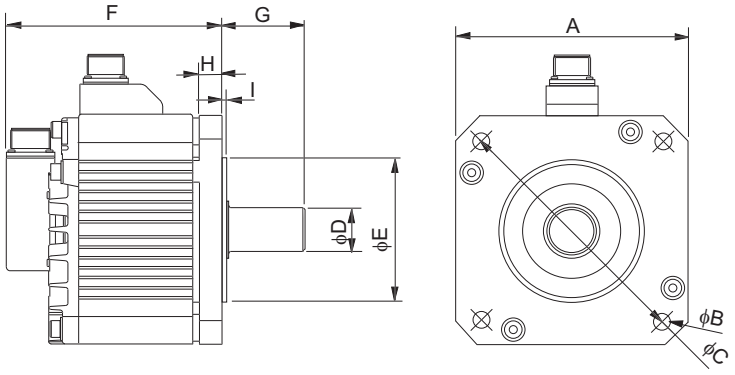
Servo motor dimensions (continued)

Low inertia models SVL-210(B)



*Sure*Servo Motor Dimensions -1000W Low Inertia	
Dimension	**SVL-210(B)**
A	100 [3.937]
B	9 [0.3543]
C	115 +0.2/-0.2 [4.258]
D	22 +0.0/-0.013 (22h6)
E	95 +0.0/-0.035 (95h7)
F (w/o brake)	158 [6.22]
F (with brake)	190 [7.48]
G	45 [1.77]
H	17 [0.669]
I	7 [0.28]
UNITS: mm [in] (Inches are for reference only; not included on diameter dimensions for accuracy.)	

Medium inertia models SVM-210(B), SVM-220(B), SVM-230(B)



*Sure*Servo Motor Dimensions -1000W-3000W Medium Inertia			
Dimension	**SVM-210(B)**	**SVM-220(B)**	**SVM-230(B)**
A	130 [5.118]	180 [7.087]	
B	9 [0.3543]	13.5 [0.5315]	
C	145 +0.2/-0.2 [5.709]	200 +0.2/-0.2 [7.874]	
D	22 +0.0/-0.013 (22h6)	35 +0.0/-0.016 (35h6)	
E	110 +0.0/-0.035 (110h7)	114.3 +0/-0.035 (114.3h7)	
F (w/o brake)	143 [5.63]	164 [6.457]	212 [8.35]
F (with brake)	181 [7.126]	213 [8.386]	258 [10.16]
G	55 [2.17]	75 [2.95]	
H	15 [0.591]	20 [0.787]	
I	4 [0.157]		
UNITS: mm [in] (Inches are for reference only; not included on diameter dimensions for accuracy.)			